

Nathan Roos

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I am a student pursuing a research Master's degree in applied mathematics for machine learning with two main-track publications (NeurIPS 2026, WACV 2026), seeking a 6-month research internship starting in April 2027 in fundamental AI or AI for science.

Publications

- D. Rizdvanetskyi, **N. Roos**, and P. Lutsik, "Data-Driven Soft Labeling Scales DNA Read Classification to Whole-Body Cell-Type Deconvolution," **NeurIPS 2026 (Main track)**
- **N. Roos**, E. Iakovleva, A. Gjergji, V.P. Pastore, and E. Tartaglione, "How I Met Your Bias: Investigating Bias Amplification in Diffusion Models," **WACV 2026 (Main track)**

Professional Experience

ML Research Intern (computational epigenomics)

KU Leuven, Dept. of Oncology, Lab. of Computational Cancer Biology and Epigenomics

 March 2026 – Aug. 2026  Leuven, Belgium

- Designed a deep learning framework (Transformers, LSTM) for cell-type deconvolution of bulk DNA samples from read-level methylation calls; paper accepted at NeurIPS 2026 (cf. Publications)
- Developed a reference-free, read-level deconvolution method: a probabilistic generative model fitted with EM (C++ implementation, custom likelihood tempering) that infers cell-type methylation profiles and proportions from multiple bulk samples

Data Scientist Intern

Banque de France (French Central Bank)

 Sept. 2025 – Feb. 2026  Paris, France

- Unsupervised detection of anomalous credit ratings using ML methods
- Development of statistical tests and Python analysis tools for the statistical testing of credit ratings

Education

MVA Master's (double degree)

ENS Paris-Saclay

 Sept. 2026 – Sept. 2027  Paris, France

Advanced courses in modern ML: geometric deep learning, optimal transport, stochastic calculus, graph and text representation learning, LLMs, and statistical/geometric data analysis

Engineering degree in Applied Mathematics / Computer Science

Télécom Paris, IP Paris

 Sept. 2023 – Sept. 2027  Paris, France

- GPA: 4/4 (Top of the class)
- Computer Science: Deep Learning (MLP, CNN, VAE, Transformers, etc.), Databases, Theory of computation, Logic, Compilation, Networking

- Applied Mathematics: Stochastic Modeling and Calculus, Numerical Analysis, Optimization, Statistics

Preparatory classes MPI*

Lycée Clémenceau

 Sept. 2021 – June 2023

 Nantes, France

- GPA: 4/4 (Top of the class)
- Placed 8th in the Mines-Ponts National Competitive Examination

Projects

Research in debiasing schemes for Deep Learning

One-year research track at Télécom Paris

 Sept. 2024 – June 2025

- Review of the state of the art in debiasing for computer vision ([link](#))
- Discovered a previously unknown impact of sampling hyperparameters on the amplification of bias by diffusion models. Resulting paper accepted at WACV 2026 (cf. Publications)

Design and deployment of a NetOps solution for an ISP

Group semester project at Télécom Paris

 Feb. 2024 – June 2024

- Jury Prize 2024 (top out of 50+ projects)
- Design and deployment of a system to automate the updating of customer router configurations for an Internet Service Provider

Optimization of work scheduling on a flow network

15-month research project (TIPE) for the competitive examinations to enter France's top engineering schools

 Feb. 2021 – April 2022

- Goal: find the work schedule on a flow network that maximizes road traffic
- Design, implementation in OCaml, and testing of four polynomial-complexity approximation algorithms
- Search for a proof of NP-hardness

Skills

ML/Deep Learning Transformers, RNN, LSTM, Diffusion models, VAE, CNN

Statistics/Probabilistic methods stochastic modeling, EM algorithms, statistical testing

Programming Python, C, C++, SQL, TypeScript/JavaScript, OCaml

Frameworks and Infrastructure PyTorch, TensorFlow, Git, Docker, CI/CD, Slurm (HPC)

Languages French (Native), English (C1), German (B1)